Since, magnitude of magnetic dipole moment is $|\vec{p}_m| = p_m = q_m \cdot 2l$ and substituting in equation (3.19), the magnetic field at a point C is

$$\vec{B}_{equatorial} = -\frac{\mu_\circ}{4\pi}\frac{p_m}{(r^2+l^2)^{\frac{3}{2}}}\hat{i} \qquad (3.20)$$

If the distance between two poles in a bar magnet is small (looks like short magnet) when compared to the distance between geometrical centre O of bar magnet and the location of point C ($r >> l$),

$$(r^2+l^2)^{\frac{3}{2}} \approx r^3 \qquad (3.21)$$

Therefore, using equation (3.21) in equation (3.20), we get

$$\vec{B}_{equatorial} = -\frac{\mu_\circ}{4\pi}\frac{p_m}{r^3}\hat{i}$$

Since $p_m\hat{i} = \vec{p}_m$, the magnetic field at equatorial point is given by

$$\vec{B}_{equatorial} = -\frac{\mu_\circ}{4\pi}\frac{\vec{p}_m}{r^3} \qquad (3.22)$$

Note that magnitude of $B_{axial}$ is twice that of magnitude of $B_{equatorial}$ and the direction of $B_{axial}$ and $B_{equatorial}$ are opposite.

## EXAMPLE 3.6

A short bar magnet has a magnetic moment of 0.5 J $T^{-1}$. Calculate magnitude and direction of the magnetic field produced by the bar magnet which is kept at a distance of 0.1 m from the centre of the bar magnet along (a) axial line of the bar magnet and (b) normal bisector of the bar magnet.

### *Solution*

Given magnetic moment = 0.5 J $T^{-1}$ and distance $r$ = 0.1 m

(a) When the point lies on the axial line of the bar magnet, the magnetic field for short magnet is given by

$$\vec{B}_{axial} = \frac{\mu_\circ}{4\pi}\left(\frac{2p_m}{r^3}\right)\hat{i}$$

$$\vec{B}_{axial} = 10^{-7} \times \left(\frac{2\times 0.5}{(0.1)^3}\right)\hat{i} = 1\times 10^{-4}\ \hat{i}\ \text{T}$$

Hence, the magnitude of the magnetic field along axial is $B_{axial} = 1 \times 10^{-4}$ T and direction is towards South to North.

(b) When the point lies on the normal bisector (equatorial) line of the bar magnet, the magnetic field for short magnet is given by

$$\vec{B}_{equatorial} = -\frac{\mu_\circ}{4\pi}\frac{p_m}{r^3}\hat{i}$$

$$\vec{B}_{equatorial} = -10^{-7}\left(\frac{0.5}{(0.1)^3}\right)\hat{i} = -0.5\times 10^{-4}\ \hat{i}\ \text{T}$$

Hence, the magnitude of the magnetic field along axial is $B_{equatorial}$ = 0.5 × $10^{-4}$ T and direction is towards North to South.

Note that magnitude of $B_{axial}$ is twice that of magnitude of $B_{equatorial}$ and the direction of $B_{axial}$ and $B_{equatorial}$ are opposite.

## 3.3 TORQUE ACTING ON A BAR MAGNET IN UNIFORM MAGNETIC FIELD

Consider a magnet of length $2l$ and pole strength $q_m$ kept in a uniform magnetic field $\vec{B}$ as shown in Figure 3.16. Each pole experiences a force of magnitude $q_mB$ but acting in opposite directions. Therefore, the net force exerted on the magnet is zero and hence, there is no translatory motion. These two equal and opposite forces constitute a couple (about midpoint of bar magnet) tend to align the magnet in the direction of the magnetic field $\vec{B}$.



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